# Certificate

## Astra Automation

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# Certificate

## Certificate overview

The Certificate App API provides CRUD (Create-Read-Update-Delete) access to certificate resources in order to represent admin-trusted certificates that are used for establishing outgoing connections.

## Create a new certificate

POST `/accounts/{account_id}/core/v1/certificates`

Indicates the desired values for the Certificate API resource to be created.

### Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| account_id | string | path | True | ID of the containing account resource<br>• example: {{.Account}} |

### Request Body

Indicates the desired values for the Certificate API resource to be created.

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | Media type of the resource. Defined values are:<br>• "application/astra-certificate"<br>• example: application/astra-certificate |
| version | string | True | Version of the resource. Defined values are:<br>• "1.0"<br>• example: 1.0 |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| certUse | string | False | JSON string indicating the use of the certificate. If not specified, defaults to "rootCA". Defined values are:<br>• "rootCA" - The certificate is a certificate authority (CA) certificate used to trust any certificate signed by that CA.<br>• "intermediateCA" - The certificate is an intermediate CA certificate issued by a root CA.<br>• example: rootCA |
| cert | string | True | JSON string containing a Base64 encoded PKCS-11 formatted certificate (PEM). Defined values are:<br>• Conforms to the base64 Encoded Data Schema<br>• example: MIIEUzCCAjugAwIBAgICEAMwDQYJKoZIhvcNAQELBXJtJWvj0fZzj1tVrA== |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| isSelfSigned | string | False | JSON string indicating whether the certificate is self-signed. If not provided on create, the value defaults to "false". On update:<br>• If the value is provided, that value will be used.<br>• If the value is not provided and the "cert" field is included in the update, the value will be set to "false".<br>• If the value is not provided and the "cert" field is not included in the update, the existing value will be retained. Defined values are:<br>• "true" - The certificate is self-signed.<br>• "false" - The certificate is not self-signed.<br>• example: false |
| trustStateDesired | string | False | JSON string containing a value indicating whether the certificate is trusted. If not specified on create, the default value is "trusted". Defined values are:<br>• "untrusted" - The certificate is not trusted.<br>• "trusted" - The certificate is marked as trusted.<br>• example: trusted |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| metadata | type_astra_metadata_update | False | Client and service-specified metadata associated with the resource. Defined values are:<br><br>• Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification. |

**Example request**

```
{
  "type": "application/astra-certificate",
  "version": "1.0",
  "certUse": "rootCA",
  "cert":
"LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSUZyVENDQTVXZ0F3MVJHbnFGbUJSSWRyV
1kwPQotLS0tLUVORCBDRVJUSUZJQ0FURS0tLS0t"
}
```

## Response

```
Status: 201, Returns the newly created certificate resource in the JSON
response body.
```

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| type | string | True | Media type of the resource. Defined values are:<br>• "application/astra-certificate"<br>• example: application/astra-certificate |
| version | string | True | Version of the resource. Defined values are:<br>• "1.0"<br>• example: 1.0 |
| id | string | True | Globally unique identifier of the resource. Defined values are:<br>• Conforms to the UUIDv4 Schema<br>• example: eb12225e-aded-584e-9da1-03881ce4cf7a |
| certUse | string | True | JSON string indicating the use of the certificate. If not specified, defaults to "rootCA". Defined values are:<br>• "rootCA" - The certificate is a certificate authority (CA) certificate used to trust any certificate signed by that CA.<br>• "intermediateCA" - The certificate is an intermediate CA certificate issued by a root CA.<br>• example: rootCA |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| cert | string | True | JSON string containing a Base64 encoded PKCS-11 formatted certificate (PEM). Defined values are:<br>• Conforms to the base64 Encoded Data Schema<br>• example: MIIEUzCCAjugAwIBAgICEAMwDQYJKoZIhvcNAQELBXJtJWvj0fZzj1tVrA== |
| cn | string | True | JSON string containing a common name extracted from the certificate. Defined values are:<br>• Minimum length: 1<br>• Maximum length: 511<br>• example: vcenter.example.com |
| expiryTimestamp | string | True | JSON string containing a timestamp indicating when the certificate expires. Defined values are:<br>• Conforms to the ISO-8601 Date Time Schema<br>• example: 2022-08-06T12:24:52.256624Z |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| isSelfSigned | string | True | JSON string indicating whether the certificate is self-signed. If not provided on create, the value defaults to "false". On update:<br>• If the value is provided, that value will be used.<br>• If the value is not provided and the "cert" field is included in the update, the value will be set to "false".<br>• If the value is not provided and the "cert" field is not included in the update, the existing value will be retained. Defined values are:<br>• "true" - The certificate is self-signed.<br>• "false" - The certificate is not self-signed.<br>• example: false |
| trustState | string | True | JSON string containing one of a set of enumerated values indicating whether the certificate is trusted. Defined values are:<br>• "untrusted" - The certificate is not trusted.<br>• "trusted" - The certificate is marked as trusted.<br>• "expired" - The certificate is expired, so is neither trusted or not trusted.<br>• example: trusted |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| trustStateTransitions | array[type_astra_stateTransitions] | True | JSON array of JSON objects, each representing the set of permitted state transitions from a given defined state. Each JSON object contains the fields "from" and "to", where the "from" value is the name of a state, and the "to" value is an array of permitted destination states. |
| trustStateDesired | string | False | JSON string containing a value indicating whether the certificate is trusted. If not specified on create, the default value is "trusted". Defined values are:<br>• "untrusted" - The certificate is not trusted.<br>• "trusted" - The certificate is marked as trusted.<br>• example: trusted |
| trustStateDetails | array[type_astra_stateDetails] | True | JSON array of JSON objects that lists additional details about the trust state of the certificate. If no details are available, this array will be empty.<br>• Minimum Items: 0 |

| Name | Type | Required | Description |
|---|---|---|---|
| metadata | type_astra_metadata_update | True | Client and service-specified metadata associated with the resource. Defined values are:<br><br>• Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification. |

**Example response**

```
{
  "type": "application/astra-certificate",
  "version": "1.0",
  "id": "eb12225e-aded-584e-9da1-03881ce4cf7a",
  "certUse": "rootCA",
  "cert":
"LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSUZyVENDQTVXZ0F3MVJHbnFGbUJSSWRyV
1kwPQotLS0tLUVORCBDRVJUSUZJQ0FURS0tLS0t",
  "cn": "vcenter.example.com",
  "isSelfSigned": "false",
  "expiryTimestamp": "2022-03-23T16:48:38.5233461Z",
  "trustState": "trusted",
  "trustStateTransitions": [
    {
      "from": "untrusted",
      "to": [
        "trusted"
      ]
    },
    {
      "from": "trusted",
      "to": [
        "untrusted"
      ]
    }
  ],
  "trustStateDesired": "trusted",
  "trustStateDetails": [],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


## Response

```
Status: 400, Bad request
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |

| Name | Type | Required | Description |
|---|---|---|---|
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |
| invalidFields | array[invalidFields] | False | List of invalid request body fields |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/7",
  "title": "Invalid JSON payload",
  "detail": "The request body is not valid JSON.",
  "status": "400"
}
```

## Response

```
Status: 403, Forbidden
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

## Response

```
Status: 500, Internal server error
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/34",
  "title": "Internal server error",
  "detail": "The server was unable to process this request.",
  "status": "500"
}
```

## Error

```
Status: 401, Unauthorized
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |

#### Example error response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

## Definitions

**See Definitions**

type_astra_label

| Name | Type | Required | Description |
|---|---|---|---|
| name | string | True | |
| value | string | True | |

type_astra_metadata_update

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

| Name | Type | Required | Description |
|---|---|---|---|
| labels | array[type_astra_label] | False | |
| creationTimestamp | string | False | |
| modificationTimestamp | string | False | |
| createdBy | string | False | |
| modifiedBy | string | False | |

type_astra_stateTransitions

| Name | Type | Required | Description |
|---|---|---|---|
| from | string | True | |
| to | array[string] | True | |

type_astra_stateDetails

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | True | |

invalidFields

| Name | Type | Required | Description |
|---|---|---|---|
| name | string | True | Name of the invalid request body field |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| reason | string | True | Reason why the request body field is invalid |

# List all certificates

GET `/accounts/{account_id}/core/v1/certificates`

Returns a JSON array of all matching resources in the collection. If individual fields are specified in the request, the items JSON array will contain the fields requested for each matching resource in the order specified.

## Parameters

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| account_id | string | path | True | ID of the containing account resource<br><br>• example: {{.Account}} |
| continue | | query | False | Indicates the token to resume listing from.<br><br>URL examples: "continue=<continue token="" from="" collection="" metadata="">"</continue><br><br>OpenAPI examples: "<continue token="" from="" collection="" metadata="">"</continue><br><br></continue></continue> |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| filter | string | query | False | Indicates which fields should be matched in order to return a resource when listing a collection.<br><br>Supported operators are:<br><br>• eq - Equality<br>• lt/gt - Less than/Greater than<br>• lte/gte - Less than or equals to/Greater than or equals to<br><br>URL example: "filter=field%20eq%20%27value%27"<br><br>OpenAPI examples: "field eq 'value'"<br><br>• example: certUse%20eq%20’rootCA' |
| include | string | query | False | Indicates which fields should be returned when listing a collection.<br><br>URL examples: "include=id", "include=id,name"<br><br>OpenAPI examples: "id"<br><br>• example: id,cn,isSelfSigned |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| limit | string | query | False | Indicates the maximum number of resources to return when listing a collection.<br><br>URL examples: "limit=2"<br><br>OpenAPI examples: "2" |
| orderBy | string | query | False | Indicates the sort order of resource when listing a collection.<br><br>URL examples: "orderBy=name", "orderBy=name%20desc"<br><br>OpenAPI examples: "name desc" |

## Response

```
Status: 200, Returns a JSON array of all matching resources in the
collection. If individual fields are specified in the request, the items
JSON array will contain the fields requested for each matching resource in
the order specified.
```

| Name | Type | Required | Description |
|---|---|---|---|
| items | array[Any of: certificate_1.0_get_response_body, array[]] | True | |
| metadata | type_astra_metadata_list | True | |

#### Example response

```
{
  "type": "application/astra-certificates",
  "version": "1.0",
  "items": [
    [
      "f97f3441-075e-4f9b-9d1e-47f9f62a3fc4",
      "prod.region-1.example.com",
      "true"
    ],
    [
      "5e365b4d-034f-4f13-95ef-d3a7fbc0df3d",
      "prod.region-2.example.com",
      "false"
    ],
    [
      "11ceb350-1999-4b79-80a6-46a980a27856",
      "prod.example.com",
      "false"
    ]
  ],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


## Response

```
Status: 401, Unauthorized
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

## Response

```
Status: 400, Bad request
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |
| invalidParams | array[invalidParams] | False | List of invalid query parameters |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

## Error

```
Status: 404, Not found
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |

| Name | Type | Required | Description |
|---|---|---|---|
| status | string | True | |
| correlationID | string | False | |

#### Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't found.",
  "status": "404"
}
```

## Definitions

**See Definitions**

type_astra_stateTransitions

| Name | Type | Required | Description |
|---|---|---|---|
| from | string | True | |
| to | array[string] | True | |

type_astra_stateDetails

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | True | |

type_astra_label

| Name | Type | Required | Description |
|---|---|---|---|
| name | string | True | |
| value | string | True | |

type_astra_metadata_update

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

| Name | Type | Required | Description |
|---|---|---|---|
| labels | array[type_astra_label] | False | |
| creationTimestamp | string | False | |
| modificationTimestamp | string | False | |
| createdBy | string | False | |
| modifiedBy | string | False | |

certificate_1.0_get_response_body

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| type | string | True | Media type of the resource. Defined values are:<br>• "application/astra-certificate"<br>• example: application/astra-certificate |
| version | string | True | Version of the resource. Defined values are:<br>• "1.0"<br>• example: 1.0 |
| id | string | True | Globally unique identifier of the resource. Defined values are:<br>• Conforms to the UUIDv4 Schema<br>• example: eb12225e-aded-584e-9da1-03881ce4cf7a |
| certUse | string | True | JSON string indicating the use of the certificate. If not specified, defaults to "rootCA". Defined values are:<br>• "rootCA" - The certificate is a certificate authority (CA) certificate used to trust any certificate signed by that CA.<br>• "intermediateCA" - The certificate is an intermediate CA certificate issued by a root CA.<br>• example: rootCA |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| cert | string | True | JSON string containing a Base64 encoded PKCS-11 formatted certificate (PEM). Defined values are:<br>• Conforms to the base64 Encoded Data Schema<br>• example: MIIEUzCCAjugAwIBAgICEAMwDQYJKoZIhvcNAQELBXJtJWvj0fZzj1tVrA== |
| cn | string | True | JSON string containing a common name extracted from the certificate. Defined values are:<br>• Minimum length: 1<br>• Maximum length: 511<br>• example: vcenter.example.com |
| expiryTimestamp | string | True | JSON string containing a timestamp indicating when the certificate expires. Defined values are:<br>• Conforms to the ISO-8601 Date Time Schema<br>• example: 2022-08-06T12:24:52.256624Z |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| isSelfSigned | string | True | JSON string indicating whether the certificate is self-signed. If not provided on create, the value defaults to "false". On update:<br>• If the value is provided, that value will be used.<br>• If the value is not provided and the "cert" field is included in the update, the value will be set to "false".<br>• If the value is not provided and the "cert" field is not included in the update, the existing value will be retained. Defined values are:<br>• "true" - The certificate is self-signed.<br>• "false" - The certificate is not self-signed.<br>• example: false |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| trustState | string | True | JSON string containing one of a set of enumerated values indicating whether the certificate is trusted. Defined values are:<br>• "untrusted" - The certificate is not trusted.<br>• "trusted" - The certificate is marked as trusted.<br>• "expired" - The certificate is expired, so is neither trusted or not trusted.<br>• example: trusted |
| trustStateTransitions | array[type_astra_stateTransitions] | True | JSON array of JSON objects, each representing the set of permitted state transitions from a given defined state. Each JSON object contains the fields "from" and "to", where the "from" value is the name of a state, and the "to" value is an array of permitted destination states. |
| trustStateDesired | string | False | JSON string containing a value indicating whether the certificate is trusted. If not specified on create, the default value is "trusted". Defined values are:<br>• "untrusted" - The certificate is not trusted.<br>• "trusted" - The certificate is marked as trusted.<br>• example: trusted |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| trustStateDetails | array[type_astra_stateDetails] | True | JSON array of JSON objects that lists additional details about the trust state of the certificate. If no details are available, this array will be empty.<br><br>• Minimum Items: 0 |
| metadata | type_astra_metadata_update | True | Client and service-specified metadata associated with the resource. Defined values are:<br><br>• Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification. |

type_astra_metadata_list

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| continue | string | False | |
| count | number | False | |
| labels | array[type_astra_label] | False | |
| creationTimestamp | string | False | |
| modificationTimestamp | string | False | |
| createdBy | string | False | |
| modifiedBy | string | False | |

invalidParams

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| name | string | True | Name of the invalid query parameter |
| reason | string | True | Reason why the query parameter is invalid |

# Retrieve a certificate

GET `/accounts/{account_id}/core/v1/certificates/{certificate_id}`

Returns the stored Certificate API resource in the JSON response body.

## Parameters

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| account_id | string | path | True | ID of the containing account resource<br>• example: {{.Account}} |
| certificate_id | string | path | True | ID of the certificate collection to list |

## Response

```
Status: 200, Returns the stored Certificate API resource in the JSON
response body.
```

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| type | string | True | Media type of the resource. Defined values are:<br>• "application/astra-certificate"<br>• example: application/astra-certificate |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| version | string | True | Version of the resource. Defined values are:<br>• "1.0"<br>• example: 1.0 |
| id | string | True | Globally unique identifier of the resource. Defined values are:<br>• Conforms to the UUIDv4 Schema<br>• example: eb12225e-aded-584e-9da1-03881ce4cf7a |
| certUse | string | True | JSON string indicating the use of the certificate. If not specified, defaults to "rootCA". Defined values are:<br>• "rootCA" - The certificate is a certificate authority (CA) certificate used to trust any certificate signed by that CA.<br>• "intermediateCA" - The certificate is an intermediate CA certificate issued by a root CA.<br>• example: rootCA |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| cert | string | True | JSON string containing a Base64 encoded PKCS-11 formatted certificate (PEM). Defined values are:<br>• Conforms to the base64 Encoded Data Schema<br>• example: MIIEUzCCAjugAwIBAgICEAMwDQYJKoZIhvcNAQELBXJtJWvj0fZzj1tVrA== |
| cn | string | True | JSON string containing a common name extracted from the certificate. Defined values are:<br>• Minimum length: 1<br>• Maximum length: 511<br>• example: vcenter.example.com |
| expiryTimestamp | string | True | JSON string containing a timestamp indicating when the certificate expires. Defined values are:<br>• Conforms to the ISO-8601 Date Time Schema<br>• example: 2022-08-06T12:24:52.256624Z |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| isSelfSigned | string | True | JSON string indicating whether the certificate is self-signed. If not provided on create, the value defaults to "false". On update:<br>• If the value is provided, that value will be used.<br>• If the value is not provided and the "cert" field is included in the update, the value will be set to "false".<br>• If the value is not provided and the "cert" field is not included in the update, the existing value will be retained. Defined values are:<br>• "true" - The certificate is self-signed.<br>• "false" - The certificate is not self-signed.<br>• example: false |
| trustState | string | True | JSON string containing one of a set of enumerated values indicating whether the certificate is trusted. Defined values are:<br>• "untrusted" - The certificate is not trusted.<br>• "trusted" - The certificate is marked as trusted.<br>• "expired" - The certificate is expired, so is neither trusted or not trusted.<br>• example: trusted |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| trustStateTransitions | array[type_astra_stateTransitions] | True | JSON array of JSON objects, each representing the set of permitted state transitions from a given defined state. Each JSON object contains the fields "from" and "to", where the "from" value is the name of a state, and the "to" value is an array of permitted destination states. |
| trustStateDesired | string | False | JSON string containing a value indicating whether the certificate is trusted. If not specified on create, the default value is "trusted". Defined values are:<br>• "untrusted" - The certificate is not trusted.<br>• "trusted" - The certificate is marked as trusted.<br>• example: trusted |
| trustStateDetails | array[type_astra_stateDetails] | True | JSON array of JSON objects that lists additional details about the trust state of the certificate. If no details are available, this array will be empty.<br>• Minimum Items: 0 |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| metadata | type_astra_metadata_update | True | Client and service-specified metadata associated with the resource. Defined values are:<br><br>• Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification. |

**Example response**

```
{
  "type": "application/astra-certificate",
  "version": "1.0",
  "id": "eb12225e-aded-584e-9da1-03881ce4cf7a",
  "certUse": "rootCA",
  "cert":
"LS0tLS1CRUdJTiBDRVJUSUZJQ0FURS0tLS0tCk1JSUZyVENDQTVXZ0F3MVJHbnFGbUJSSWRyV
1kwPQotLS0tLUVORCBDRVJUSUZJQ0FURS0tLS0t",
  "cn": "vcenter.example.com",
  "expiryTimestamp": "2022-03-23T16:48:38.5233461Z",
  "isSelfSigned": "true",
  "trustState": "untrusted",
  "trustStateTransitions": [
    {
      "from": "untrusted",
      "to": [
        "trusted"
      ]
    },
    {
      "from": "trusted",
      "to": [
        "untrusted"
      ]
    }
  ],
  "trustStateDetails": [],
  "metadata": {
    "labels": [],
    "creationTimestamp": "2022-10-06T20:58:16.305662Z",
    "modificationTimestamp": "2022-10-06T20:58:16.305662Z",
    "createdBy": "8f84cf09-8036-51e4-b579-bd30cb07b269"
  }
}
```


## Response

```
Status: 401, Unauthorized
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |

| Name | Type | Required | Description |
|---|---|---|---|
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

## Response

```
Status: 400, Bad request
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |
| invalidParams | array[invalidParams] | False | List of invalid query parameters |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

## Response

```
Status: 403, Forbidden
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

## Error

```
Status: 404, Not found
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |

#### Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't found.",
  "status": "404"
}
```

## Definitions

**See Definitions**

type_astra_stateTransitions

| Name | Type | Required | Description |
|---|---|---|---|
| from | string | True | |
| to | array[string] | True | |

type_astra_stateDetails

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | True | |

type_astra_label

| Name | Type | Required | Description |
|---|---|---|---|
| name | string | True | |
| value | string | True | |

type_astra_metadata_update

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

| Name | Type | Required | Description |
|---|---|---|---|
| labels | array[type_astra_label] | False | |
| creationTimestamp | string | False | |
| modificationTimestamp | string | False | |
| createdBy | string | False | |
| modifiedBy | string | False | |

invalidParams

| Name | Type | Required | Description |
|---|---|---|---|
| name | string | True | Name of the invalid query parameter |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| reason | string | True | Reason why the query parameter is invalid |

# Modify a certificate

PUT `/accounts/{account_id}/core/v1/certificates/{certificate_id}`

Replaces the entire stored resource with the supplied JSON request body.

**Side Effects**

- An updated CA certificate will be updated in the trust store.
- If an existing trusted CA certificate becomes untrusted, it will be removed from the trust store.
- If a certificate body is updated ("cert" field), the "isSelfSigned" indicator must be included to indicate the state of the new certificate. If the "isSelfSigned" indicator is not included in this case, the value will be set to "false".
- If an update does not modify the certificate body ("cert" field), the "isSelfSigned" indicator is optional and the existing value will be retained if not provided.

## Parameters

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| account_id | string | path | True | ID of the containing account resource<br>• example: {{.Account}} |
| certificate_id | string | path | True | ID of the certificate collection to list |

## Request Body

Replaces the entire stored resource with the supplied JSON request body.

**Side Effects**

- An updated CA certificate will be updated in the trust store.
- If an existing trusted CA certificate becomes untrusted, it will be removed from the trust store.
- If a certificate body is updated ("cert" field), the "isSelfSigned" indicator must be included to indicate the state of the new certificate. If the "isSelfSigned" indicator is not included in this case, the value will be set to "false".
- If an update does not modify the certificate body ("cert" field), the "isSelfSigned" indicator is optional and the existing value will be retained if not provided.

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| type | string | True | Media type of the resource. Defined values are:<br>• "application/astra-certificate"<br>• example: application/astra-certificate |
| version | string | True | Version of the resource. Defined values are:<br>• "1.0"<br>• example: 1.0 |
| id | string | False | Globally unique identifier of the resource. Defined values are:<br>• Conforms to the UUIDv4 Schema<br>• example: eb12225e-aded-584e-9da1-03881ce4cf7a |
| certUse | string | False | JSON string indicating the use of the certificate. If not specified, defaults to "rootCA". Defined values are:<br>• "rootCA" - The certificate is a certificate authority (CA) certificate used to trust any certificate signed by that CA.<br>• "intermediateCA" - The certificate is an intermediate CA certificate issued by a root CA.<br>• example: rootCA |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| cert | string | False | JSON string containing a Base64 encoded PKCS-11 formatted certificate (PEM). Defined values are:<br>• Conforms to the base64 Encoded Data Schema<br>• example: MIIEUzCCAjugAwIBAgICEAMwDQYJKoZIhvcNAQELBXJtJWvj0fZzj1tVrA== |
| cn | string | False | JSON string containing a common name extracted from the certificate. Defined values are:<br>• Minimum length: 1<br>• Maximum length: 511<br>• example: vcenter.example.com |
| expiryTimestamp | string | False | JSON string containing a timestamp indicating when the certificate expires. Defined values are:<br>• Conforms to the ISO-8601 Date Time Schema<br>• example: 2022-08-06T12:24:52.256624Z |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| isSelfSigned | string | False | JSON string indicating whether the certificate is self-signed. If not provided on create, the value defaults to "false". On update:<br>• If the value is provided, that value will be used.<br>• If the value is not provided and the "cert" field is included in the update, the value will be set to "false".<br>• If the value is not provided and the "cert" field is not included in the update, the existing value will be retained. Defined values are:<br>• "true" - The certificate is self-signed.<br>• "false" - The certificate is not self-signed.<br>• example: false |
| trustState | string | False | JSON string containing one of a set of enumerated values indicating whether the certificate is trusted. Defined values are:<br>• "untrusted" - The certificate is not trusted.<br>• "trusted" - The certificate is marked as trusted.<br>• "expired" - The certificate is expired, so is neither trusted or not trusted.<br>• example: trusted |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| trustStateTransitions | array[type_astra_stateTransitions] | False | JSON array of JSON objects, each representing the set of permitted state transitions from a given defined state. Each JSON object contains the fields "from" and "to", where the "from" value is the name of a state, and the "to" value is an array of permitted destination states. |
| trustStateDesired | string | False | JSON string containing a value indicating whether the certificate is trusted. If not specified on create, the default value is "trusted". Defined values are:<br>• "untrusted" - The certificate is not trusted.<br>• "trusted" - The certificate is marked as trusted.<br>• example: trusted |
| trustStateDetails | array[type_astra_stateDetails] | False | JSON array of JSON objects that lists additional details about the trust state of the certificate. If no details are available, this array will be empty.<br>• Minimum Items: 0 |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| metadata | type_astra_metadata_update | False | Client and service-specified metadata associated with the resource. Defined values are:<br><br>• Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification. |

**Example request**

```
{
  "type": "application/astra-certificate",
  "version": "1.0",
  "trustStateDesired": "trusted"
}
```

## Response

```
Status: 204, Indicates if the certificate resource was updated.
```

## Response

```
Status: 401, Unauthorized
```

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |

| Name | Type | Required | Description |
|---|---|---|---|
| correlationID | string | False | |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/3",
  "title": "Missing bearer token",
  "detail": "The request is missing the required bearer token.",
  "status": "401"
}
```

## Response

```
Status: 400, Bad request
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |
| invalidFields | array[invalidFields] | False | List of invalid request body fields |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/5",
  "title": "Invalid query parameters",
  "detail": "The supplied query parameters are invalid.",
  "status": "400"
}
```

## Response

```
Status: 409, Conflict
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |
| invalidFields | array[invalidFields] | False | List of invalid request body fields |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/10",
  "title": "JSON resource conflict",
  "detail": "The request body JSON contains a field that conflicts with an idempotent value.",
  "status": "409"
}
```

## Response

```
Status: 403, Forbidden
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |

#### Example response

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

## Error

```
Status: 404, Not found
```

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |

#### Example error response

```
{
  "type": "https://astra.netapp.io/problems/2",
  "title": "Collection not found",
  "detail": "The collection specified in the request URI wasn't found.",
  "status": "404"
}
```

## Definitions

**See Definitions**

type_astra_stateTransitions

| Name | Type | Required | Description |
|---|---|---|---|
| from | string | True | |
| to | array[string] | True | |

type_astra_stateDetails

| Name | Type | Required | Description |
|---|---|---|---|
| type | string | True | |
| title | string | True | |
| detail | string | True | |

type_astra_label

| Name | Type | Required | Description |
|---|---|---|---|
| name | string | True | |
| value | string | True | |

type_astra_metadata_update

Client and service-specified metadata associated with the resource. Defined values are:

- Conforms to the Astra Metadata Schema If not specified on create, a metadata object will be created with no labels. If not specified on update, the metadata object's labels, creationTimestamp and createdBy, will be preserved without modification.

| Name | Type | Required | Description |
|---|---|---|---|
| labels | array[type_astra_label] | False | |
| creationTimestamp | string | False | |
| modificationTimestamp | string | False | |
| createdBy | string | False | |
| modifiedBy | string | False | |

invalidParams

| Name | Type | Required | Description |
|---|---|---|---|
| name | string | True | Name of the invalid query parameter |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| reason | string | True | Reason why the query parameter is invalid |

invalidFields

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| name | string | True | Name of the invalid request body field |
| reason | string | True | Reason why the request body field is invalid |

# Delete a certificate

DELETE `/accounts/{account_id}/core/v1/certificates/{certificate_id}`

Removes the specified Certificate API resource.

## Parameters

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| account_id | string | path | True | ID of the containing account resource<br>• example: {{.Account}} |
| certificate_id | string | path | True | ID of the certificate collection to list |

## Response

```
Status: 204, Indicates if the certificate resource was deleted.
```

## Error

```
Status: 403, Forbidden
```

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| type | string | True | |

| Name | Type | Required | Description |
| --- | --- | --- | --- |
| title | string | True | |
| detail | string | False | |
| status | string | True | |
| correlationID | string | False | |

**Example error response**

```
{
  "type": "https://astra.netapp.io/problems/11",
  "title": "Operation not permitted",
  "detail": "The requested operation isn't permitted.",
  "status": "403"
}
```

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